LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Wiring bend radius requirements for various installations.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Wiring bend radius requirements for various installations.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of correct bending techniques on sample wire; Safety: Avoid sharp tools; Supervision: Instructor monitors hand placement.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of aircraft wiring degradation?
(b) Why is maintaining the correct bend radius critical for wire integrity?
(c) Can you list three common locations where wiring installations occur?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Minimum bend radius definitions • Impact of sharp bends on insulation • Conductor fatigue and breakage risks • Standards for rigid vs flexible wiring • Consequences of exceeding limits	L & Ds	PPT, IB	15 min
(b)	• Installation techniques for tight spaces • Use of bending tools and jigs • Handling multi-conductor cables • Preventing kinks during routing • Verification of final installation	L & Ds	PPT, PDF	20 min
(c)	• Inspection criteria for installed wiring • Documentation of bend radius compliance • Troubleshooting common installation errors • Maintenance checks for wire integrity • Reporting deviations to maintenance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a bend radius smaller than the minimum requirement affects the wire's lifespan.
(b) Describe the correct procedure for routing a wire around a tight corner without damaging the insulation.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of wiring bend radius apply to the marking processes discussed in LP 14.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Common manufacturer marking process.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min